

ПРИЛОЖЕНИЕ 3 (справочное) ДЕМУЛЬТИПЛЕКСОР

27окт2021

<https://pdf1.alldatasheet.com/datasheet-pdf/view/51038/FAIRCHILD/74138.html>

August 1986
Revised March 2000

DM74LS138 • DM74LS139 Decoder/Demultiplexer

General Description

These Schottky-clamped circuits are designed to be used in high-performance memory-decoding or data-routing applications, requiring very short propagation delay times. In high-performance memory systems these decoders can be used to minimize the effects of system decoding. When used with high-speed memories, the delay times of these decoders are usually less than the typical access time of the memory. This means that the effective system delay introduced by the decoder is negligible.

The DM74LS138 decodes one-of-eight lines, based upon the conditions at the three binary select inputs and the three enable inputs. Two active-low and one active-high enable inputs reduce the need for external gates or inverters when expanding. A 24-line decoder can be implemented with no external inverters, and a 32-line decoder requires only one inverter. An enable input can be used as a data input for demultiplexing applications.

The DM74LS139 comprises two separate two-line-to-four-line decoders in a single package. The active-low enable input can be used as a data line in demultiplexing applications.

All of these decoders/demultiplexers feature fully buffered inputs, presenting only one normalized load to its driving circuit. All inputs are clamped with high-performance Schottky diodes to suppress line-ringing and simplify system design.

Features

- Designed specifically for high speed:
 - Memory decoders
 - Data transmission systems
- DM74LS138 3-to-8-line decoders incorporates 3 enable inputs to simplify cascading and/or data reception
- DM74LS139 contains two fully independent 2-to-4-line decoders/demultiplexers
- Schottky clamped for high performance
- Typical propagation delay (3 levels of logic)

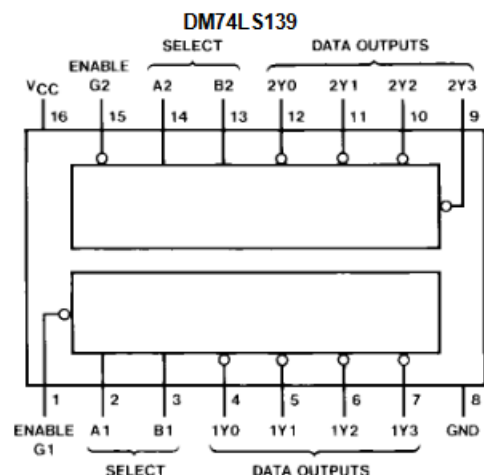
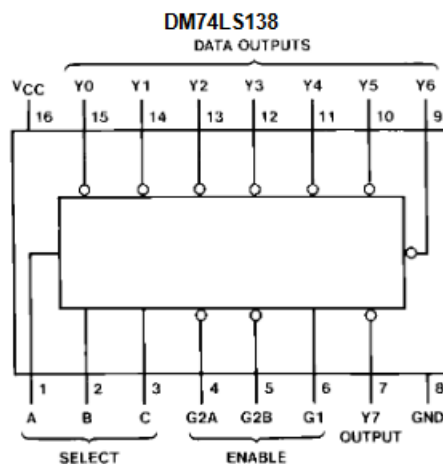
DM74LS138	21 ns
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- Typical power dissipation

DM74LS138	32 mW
DM74LS139	34 mW

DM74LS138 • DM74LS139 Decoder/Demultiplexer

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Connection Diagrams



Function Tables

DM74LS138

Inputs			Outputs							
Enable		Select								
G1	G2 (Note 1)	C B A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X X X	H	H	H	H	H	H	H	H
L	X	X X X	H	H	H	H	H	H	H	H
H	L	L L L	L	H	H	H	H	H	H	H
H	L	L L H	L	H	H	H	H	H	H	H
H	L	L H L	L	H	H	L	H	H	H	H
H	L	L H H	L	H	H	H	L	H	H	H
H	L	H L L	L	H	H	H	H	L	H	H
H	L	H L H	L	H	H	H	H	L	H	H
H	L	H H L	L	H	H	H	H	H	L	H
H	L	H H H	L	H	H	H	H	H	H	L

DM74LS139

Inputs			Outputs			
Enable		Select				
G	B	A	Y0	Y1	Y2	Y3
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	L	H	H	L	H	H
L	H	L	H	H	L	H
L	H	H	H	H	H	L

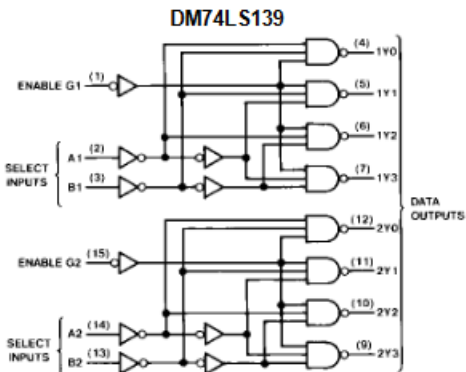
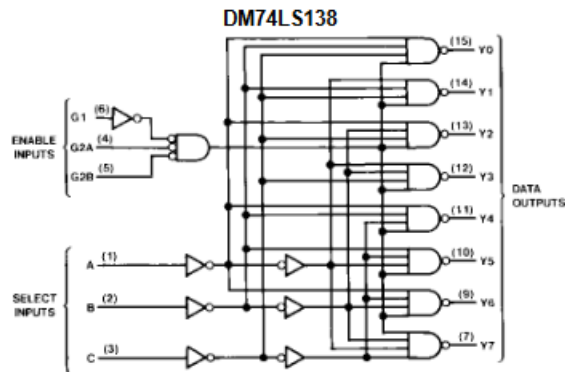
H = HIGH Level

L = LOW Level

X = Don't Care

Note 1: G2 = G2A + G2B

Logic Diagrams



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Absolute Maximum Ratings (Note 2)

Supply Voltage	7V
Input Voltage	7V
Operating Free Air Temperature Range	0°C to +70°C
Storage Temperature Range	-65°C to +150°C

Note 2: The "Absolute Maximum Ratings" are those values beyond which the safety of the device cannot be guaranteed. The device should not be operated at these limits. The parametric values defined in the Electrical Characteristics tables are not guaranteed at the absolute maximum ratings. The "Recommended Operating Conditions" table will define the conditions for actual device operation.

DM74LS138 Recommended Operating Conditions

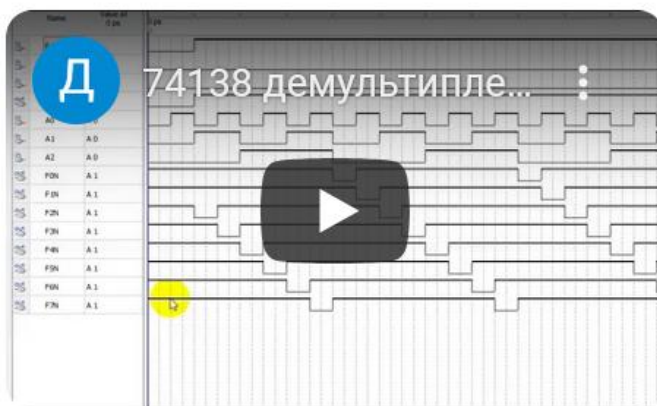
Symbol	Parameter	Min	Nom	Max	Units
V _{CC}	Supply Voltage	4.75	5	5.25	V
V _{IH}	HIGH Level Input Voltage	2			V
V _{IL}	LOW Level Input Voltage			0.8	V
I _{OH}	HIGH Level Output Current			-0.4	mA
I _{OL}	LOW Level Output Current			8	mA
T _A	Free Air Operating Temperature	0		70	°C

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DM74LS138 Electrical Characteristics

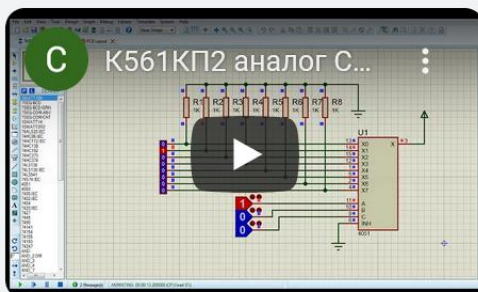
over recommended operating free air temperature range (unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ (Note 3)	Max	Units
V_I	Input Clamp Voltage	$V_{CC} = \text{Min}, I_I = -18 \text{ mA}$			-1.5	V
V_{OH}	HIGH Level Output Voltage	$V_{CC} = \text{Min}, I_{OH} = \text{Max}, V_{IL} = \text{Max}, V_{IH} = \text{Min}$	2.7	3.4		V
V_{OL}	LOW Level Output Voltage	$V_{CC} = \text{Min}, I_{OL} = \text{Max}, V_{IL} = \text{Max}, V_{IH} = \text{Min}$		0.35	0.5	V
		$I_{OL} = 4 \text{ mA}, V_{CC} = \text{Min}$		0.25	0.4	
I_I	Input Current @ Max Input Voltage	$V_{CC} = \text{Max}, V_I = 7V$			0.1	mA
I_{IH}	HIGH Level Input Current	$V_{CC} = \text{Max}, V_I = 2.7V$			20	μA
I_{IL}	LOW Level Input Current	$V_{CC} = \text{Max}, V_I = 0.4V$			-0.36	mA
I_{OS}	Short Circuit Output Current	$V_{CC} = \text{Max}$ (Note 4)	-20		-100	mA
I_{CC}	Supply Current	$V_{CC} = \text{Max}$ (Note 5)		6.3	10	mA

Note 3: All typicals are at $V_{CC} = 5V$, $T_A = 25^\circ\text{C}$.**Note 4:** Not more than one output should be shorted at a time, and the duration should not exceed one second.**Note 5:** I_{CC} is measured with all outputs enabled and OPEN.**74138 демультиплексор 1 в 8**

YouTube

05:27 · 11 февраля

**К561кп2 аналог Cd4051а в режиме демультиплексора**

atmel avr (computer processor), avr, mcu, microcontroller (computer processor), proteus, proteus flowcode isis ares, урок, flowcode, isis, ares, hiasm, arduino, микроконтроллер, программирование...

YouTube

04:39 · 6 марта